

Potential Impacts of Sand Mining on Hydrodynamics and Fine Sediment Suspension and Deposition on an Inner-shelf Shoal

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ABSTRACT

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Wave and hydrodynamic models are implemented to investigate whether sand mining on a transgressive sand shoal could influence hydrodynamic processes and sedimentary environment of the shoal covered with fine sediments. An effort is made to evaluating shoal hydrodynamic response to a hypothetical dredging scenario under a host of probable storm conditions. The model results show that changes in hydrodynamic and sedimentary environment strongly depend on the storm conditions and the shoal bathymetry. Wave dissipation, flow conditions and sediment re-suspension are significantly higher on the shallower western shoal and gradually decrease toward the deeper eastern section of the shoal. The finding suggests that the deeper eastern shoal can be exposed to frequent deposition of fine sediments, as opposed to the western shoal, despite its location farther away from the nearby major river mouth. It is concluded that large-scale sand mining, in particular from the shallower western shoal, is expected to profoundly alter the hydrodynamics and sediment suspension and deposition processes, which may also influence the benthic ecology of the shoal. The mining from the deeper eastern shoal may result in lesser impacts than the mining from the ecologically sensitive western shoal.

ADDITIONAL INDEX WORDS: Numerical model, winter storm, hurricane, muddy sediment, low-energy coast, sub-tropical, Ship Shoal, Louisiana.

INTRODUCTION

Irregular bottom topography in shallow waters such as sand shoals has been known to influence coastal hydrodynamics and bottom boundary layer dynamics (Pepper and Stone, 2004; Stone and Xu, 1996). Those sand bodies have been recognized as a viable sand resource for restoring rapidly deteriorating barrier islands and beaches along the northern Gulf coasts (Byrnes *et al.*, 1999; Khalil *et al.*, 2007; Maa *et al.*, 2004; Michel *et al.*, 2001). Waves and hydrodynamics over an inner-shelf shoal bathymetry are more complicated than other parts of the inner shelf. For instance, such bathymetric highs act as submerged breakwaters, mitigating wave energy and influence wave refraction, flow patterns, and consequent sediment transport processes (Jose *et al.*, 2007; Pepper and Stone, 2004; Stone and Xu, 1996).

There are a growing number of publications which examine changes in hydrodynamics associated with sand mining, particularly waves and their impacts on longshore transport along beaches and barriers (cf. Byrnes *et al.*, 2004; Stone and Xu, 1996). There are studies on the potential impacts of morpho-hydrodynamics, too. A notable one is the SANDPIT project,

which is a large-scale project conducted in Europe dealing primarily with potential impacts of sand mining on physical processes (Van Rijn *et al.*, 2005). In the US, most of the studies in the subject have focused on potential impacts of sand mining on wave transformation (Maa *et al.*, 2004) and benthic habitats (Palmer *et al.*, 2008), and few have been discussed regarding the modification in hydrodynamics and their potential implications for sedimentary environment of shallow sand bodies as a consequence of deepening bathymetry due to sand removal for coastal restoration.

Ship Shoal located approximately 65 km southeast of the Atchafalaya River mouth is the largest sand body off Louisiana (Figure 1). It has been recently recognized as being unique from a sediment dynamic perspective in addition to its biological habitat (Condrey and Gelpi, 2008; Grippio *et al.*, 2009; Kobashi *et al.*, 2007). Repeated field measurements from the shoal indicated that the shoal, which was previously recognized as a sandy bottom, is covered with fluvially derived fine sediments (Kobashi, 2009). The fine sediments originated primarily from the Atchafalaya River and the adjoining bay are transported to the shoal due to a shift in storm winds, which push sediment plume from the river to the shoal (Kobashi, 2009). The shoal has been recognized as a potential sand borrow site to rebuild rapidly deteriorating barrier islands (see Table 1), which act as the first line of defense against hurricane-induced storm surge and waves (Penland *et al.*, 2005; Stone *et al.*, 2005). Though

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sand replenishment is considered a viable option to restore the disintegrating barrier islands, sand mining may cause a negative impact on the physical and biological environment of the shoal. Therefore, understanding the salient morpho-hydrodynamics of the shoal plays a crucial role in assessing the impacts, given the lack of available scientific literature.

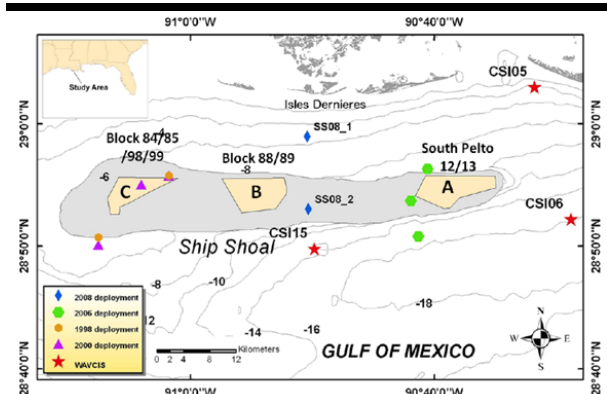


Figure 1. Study area along northern Gulf of Mexico. Gray shaded area is Ship Shoal. Shaded polygons A, B and C are the proposed mining areas.

This paper discusses potential impacts of sand mining on hydrodynamics and sediment dynamics by means of a numerical approach. The focus is on how fine sediments covered on the shoal could be influenced by the changes in the shoal bathymetry associated with sand mining. We conducted numerical model experiment for a hypothetical mining scenario and compared morpho-hydrodynamics with two bathymetric configurations and a host of probable storm conditions. The comparison enables us to understand how changes in the shoal bathymetry could influence morpho-hydrodynamics and consequently, identify potential impacts associated with those changes. The results and discussion focus on wave transformation, current variability and sediment suspension, deposition and transport as a result of varying wave and wind conditions along with modified bathymetries.

ENVIRONMENTAL SETTINGS

The Louisiana-Texas (LATEX) Shelf is located along the

the region. The Mississippi and Atchafalaya rivers discharge $15,098 \text{ m}^3/\text{s}$ and $5,569 \text{ m}^3/\text{s}$ of freshwater on average between 1993 and 2015, respectively. Sediment discharge from both rivers to the receiving basin is approximately 130×10^6 tons/year and 80×10^6 tons/year, respectively (Milliman and Meade, 1983). The sediments, which consist primarily of silt and clay, discharge from both rivers and move westward following the prevailing coastal current (Wells and Kemp, 1981). A large quantity of fine sediments discharged from the rivers create predominantly muddy bottoms on the shelf. The coast is also known for its remnant paleo-barrier islands, which have been reworked by wave action and transformed into transgressive sand shoals, one of which is Ship Shoal (Roberts, 1997). The region experiences quasi-periodic passages (every 3-10 days) of cold fronts between October and May and occasional landfalls of tropical cyclones, aka hurricanes (DiMego *et al.*, 1976).

Wind and wave-climates are characterized by low-energy environment (e.g., Georgiou *et al.*, 2005). The *in-situ* data from WACIS CSI-15 (see Figure 1 for the location) for the period between 2007/01/01 and 2008/10/01 shows that for approximately 89 percent of time (568.7 days), prevailing wind speed is less than 10 m s^{-1} , 11 percent (70.3 days) between 10 and 20 m s^{-1} , 0.13 percent (8.3 days) above 20 m s^{-1} (severe weather). Wind predominantly blows from the northeast and southeast quadrant (39%). Significant wave height (hereafter wave height) is mostly less than 1.0 m (68.3%) and only 2.0 percent of the wave height exceeds 1.5 m. Wave direction is mostly from between the east and south quadrant (67.8%). Wind direction from northern quadrant is largely associated with the passage of cold fronts and to some extent with the sea and land breezes (Hsu, 1998; Stone *et al.*, 2005).

Coastal currents follow seasonal wind patterns: eastward (up-coast) during summer and westward (down-coast) during non-summer seasons (Cochrane and Kelly, 1986). When winter storms (*i.e.* cold fronts) pass over the region, strong northerly/north-easterly winds drive the current to the east/southeast to offshore. Tidal range is less than 1 meter and tidal current is weak (a few cm s^{-1} except for the region west of the Atchafalaya Bay, cf DiMarco and Reid, 1998).

METHODS

In order to quantify potential impacts of sand mining on shoal morpho-hydrodynamics, a common approach is to implement a

Table 1. A list of the barrier islands proposed for restoration. Caminada headland is located roughly 40 km east of CSI-5 in Figure 1. Isles Dernieres is a barrier island chain and consists of several barrier islands (See Figure 1).

Area	Restoration Area	Rate of Shoreline Change (m/year)	Volume Needed for Restoration (m^3)
Caminada headland	Caminada headland	-2.6 ^{*1}	$6.1\text{-}7.6 \times 10^6$
Whisley Island	Isles Dernieres	-27.1 ^{*1}	3.1×10^6 ^{*2}
Trinity Island	Isles Dernieres	-19.1 ^{*1}	3.1×10^6 ^{*2}
Entire Isles Dernieres	Isles Dernieres	-18.9 ^{*1}	11.6×10^6 ^{*3}

Source: Penland *et al.* (2005) ^{*1}, Khalil *et al.* (2007) ^{*2}, van Heerden *et al.* (1993) ^{*3}

northwestern Gulf of Mexico. It is a broad, low-energy shelf. The Mississippi River and its distributary, the Atchafalaya River, account for over 90 percent of the total river discharge of

coupled hydrodynamic and morphological model. However, numerical simulation of morphodynamics by implementing a fully coupled 3-D hydrodynamic-morphological model in a

complex environmental setting such as our study region is highly complicated. Lack of field data with enough spatial and temporal coverage makes the model validation and analysis even more challenging. We chose a different modeling approach, which is to implement non-coupled wave and hydrodynamic models with two different bathymetry configurations and subsequently evaluate sediment suspension, deposition and transport for different sections of the shoal with the model outputs and an index that we developed. We believe that this approach enables us to gain fundamental understanding of shoal morpho-hydrodynamic processes associated with sand mining because wave and hydrodynamic processes play a dominant role in sediment transport. Therefore, wave and hydrodynamic models are implemented separately. This approach is thought to be more suitable for evaluating the impacts than the implementation of complex coupled models of wave, hydrodynamic, and sediment transport when the information necessary to validate and analyze such model output is insufficient. That being said, there are limitations to this approach: I) since the models are not a coupled system, wave-current interactions cannot be discussed and II) the evaluation of sediment transport processes is rather qualitative than quantitative. Nevertheless, the conclusions derived from the model result should be robust as we were able to identify salient morpho-hydrodynamic processes linking to sand mining. Even though the model is a semi-idealized model, the model result has been corroborated with the findings from the field measurements undertaken in the past (Kobashi, 2009).

Model Experiment

A third-generation spectral wave model, MIKE21 SW (hereafter SW) and a three-dimensional hydrodynamic model, MIKE3 HD (hereafter HD) are implemented in this study. Both models have been developed by DHI™ Water and Environment (DHI, 2005a; 2005b). The SW model has been successfully implemented for the Gulf of Mexico and the Louisiana Shelf (Jose *et al.*, 2007; Jose and Stone, 2006), as part of a wave forecasting study. The HD model has been implemented for different coastal modeling studies around the world (Freeman *et al.*, 2015; Pietrzak *et al.*, 2002; Zhang 2006). However, this model has the distinction of being implemented for the first time for evaluating the hydrodynamics of transgressive sand shoals off the Louisiana Coast. Both models are validated against in-situ data prior to the model implementation (Kobashi, 2009). The detailed model configuration including the computational domain and input parameters as well as initial conditions is briefly described in the following sub-sections.

Model Case Study

A case study (case A) is conducted to numerically compare the

changes in hydrodynamics of the region corresponding to two contrasting bathymetric configurations: one with the shoal present (Figure 2 top) and the other with the shoal completely removed (Figure 2 bottom). It should be noted that complete sand removal from the shoal would not be a realistic mining scenario for barrier island restoration even if the impacts were minimal, given the fact that there are numerous oil and natural gas pipelines that crisscross the shoal. However, this extreme hypothetical comparison would provide an excellent opportunity to quantify potential impacts of the shoal on the regional hydrodynamics.

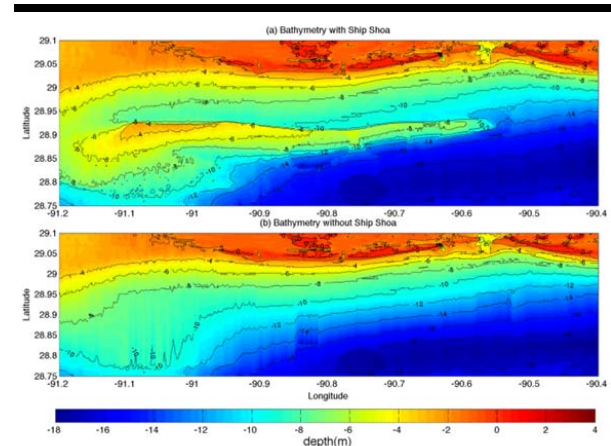


Figure 2. Bathymetry for case studies: (Top) With shoal and (bottom) without shoal.

Different deep-water wave conditions for the Louisiana Shelf are selected for the case study, following Stone and Xu (1996) (Table 2). Also, five plausible wind conditions, namely, severe storms (case A1), strong storms (case A2), moderate storms (case A3), weak storms (case A4), and fair weather (case A5) are selected (see Table 2). Stone and Xu (1996) concluded from their case study for the Louisiana Coast, that the waves from the southeast (*i.e.* swells) along with predominant southeasterly winds yielded significant changes in wave refraction and had the highest energy dissipation rates. In this study, offshore wave boundary conditions are selected as southeasterly waves (*i.e.* 135 degrees), applied uniformly along the southern boundary (Table 2). Also, the model is forced with a constant wind field in time and space over the domain and repeated with different wind speeds and directions listed in Table 2. In general, for the Louisiana Coast, currents in the shallow inner shelf and shoal are highly influenced by prevailing wind forcing, in particular during storms, given the fact that tidal currents are generally weak on the Louisiana Shelf (Wright *et al.*, 1997). In this study,

Table 2. Wind and wave conditions for case A.

Case	Wind speed	Wind direction	H_s	T_p	Wave direction
A1	20	NE(45), SE(135), SW(225), NW(315)	6	11	135
A2	15	NE(45), SE(135), SW(225), NW(315)	4	9	135
A3	12	NE(45), SE(135), SW(225), NW(315)	3	7	135
A4	10	NE(45), SE(135), SW(225), NW(315)	2	6	135
A5	5	NE(45), SE(135), SW(225), NW(315)	1	5	135

Units: $m\ s^{-1}$ (wind speed), degrees (Wave and wind directions), m (wave height, H_s), and second (wave period, T_p)

four wind directions are selected based on long-term wind and wave climate data analysis (Table 2).

Model Domain and Configuration

The model domain encompasses the entire Ship Shoal, south of the Isles Dernieres barrier island chain, and also includes three ocean observing stations under the WAVCIS network (Figure 3). Of the two bathymetries used for the model, the bathymetry without the shoal is created based on the linear interpolation between the north and south edges of the shoal defined as the 10-m isobaths, as if the shoal has been completely removed. An unstructured triangular mesh grid is developed with an embedded high-resolution mesh grid encompassing the shoal and its boundaries (Figure 3). The mesh size is based on the area of triangular grid element with maximal size of 2.0×10^{-5} degree² over the shoal and 2.0×10^{-4} degree² for the surrounding area. For the HD, offshore mesh size is selected as 2.0×10^{-3} degree².

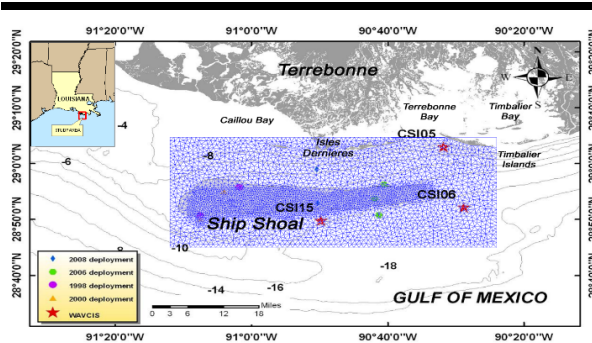


Figure 3. Model domain and mesh grids. Marks represent different observing stations.

For the case study, deep-water wave boundary conditions are applied along the southern boundary and other open boundaries (East and West) are treated as radiative boundaries for the SW model (Table 2). The HD model had all four boundaries treated as closed boundaries. In order to avoid vortex effects near the closed boundaries, the computational domain for the HD model is expanded much larger than the domain for the wave model. The flow dynamics of the region is primarily wind-driven as tides are micro-tidal. If the current driven by the boundary conditions influences the area of our interest (*i.e.* shoal), we should have seen it from the result, which was not the case. We only consider the same area as the wave model domain to discuss current variability over the shoal for the sake of comparison.

Input Parameters and Initial Conditions

Input data are collected from various data sources. Bathymetry data from the NGDC (National Geophysical Data Center) coastal relief model (Divins and Metzger, 2008) are used. The bottom friction for the SW model is estimated from a constant Nikuradse roughness height of 0.04 m. For the HD, the bottom friction is estimated from quadratic stress law, using bottom drag coefficient. Density changes (*e.g.*, stratification) are not included in the HD (*i.e.* barotropic model). Also, the

effects of wave fields on the coastal currents are not included. The time step is selected as 150 and 10 seconds for SW and HD, respectively. The model is run until the model reaches the stationary state.

Model Outputs

For the data analysis, the simulated waves and currents are extracted for various parts of the shoal. We regard them as west, middle, east, and outside (far field location) in this manuscript. The locations of each are listed in Table 3.

Table 3. Extracted locations of the model output.

Location	Longitude	Latitude	Depth(m)
West	-91.05° W	28.90° N	4.5
Middle	-90.85° W	28.90° N	6.0
East	-90.65° W	28.90° N	9.0
Outside	-90.80° W	28.85° N	16.0

Sediment Re-suspension Intensity (RI)

Changes in sediment suspension and deposition influence sediment transport rates and consequent bed elevations (Soulsby, 1997; Whitehouse *et al.*, 2000). However, it is difficult to calibrate a sediment transport model for a mixture of sand and mud bottoms primarily due to lack of field data to validate the model. Our focus on sediment dynamics is to investigate if sand mining could influence re-suspension and deposition of fluvially derived fine sediments, which potentially has a profound impact on benthic habitats of the shoal. Here we propose an alternative, effective way to estimate the potential for sediment suspension and deposition. We developed an index (Resuspension intensity: RI) to estimate intensity of sediment re-suspension from the computed bulk wave parameters. The re-suspension intensity (RI) is defined as wave-induced shear stress subtracted by critical shear stress for sediment suspension (*i.e.* excess shear stress) (Equation 1):

$$RI = \tau_{Wave} - \tau_{crit} \quad (1)$$

The positive values represent re-suspension and the negative values represent deposition. The reason why we chose *wave-induced* shear stress is because the sediment suspension is caused primarily by waves, and not by current itself for the study region (Kobashi and Jose, unpublished manuscript). Wave shear stress is estimated from Madsen (1976) and the critical shear stress for fine sediments (*i.e.* silt and clay) is chosen as a constant value of $0.15 \text{ (N m}^{-2}\text{)}$ with reference to Wright *et al.* (1997). The value applies to cohesive sediments since our focus is on potential impacts of fine sediments on the mining, although the shoal face consists of a mixture of sand and mud (mud covered on top of sandy bottom). It is expected that the high RI leads to high suspended sediment concentration (SSC) (Whitehouse *et al.*, 2000). Though the relationship is non-linear, the RI can be used as a proxy for SSC. In fact, excess shear stress and SSC had a positive correlation when comparing the in-situ data from a location near the shoal and roughly 50 km from the study region (Kobashi, 2009). The combination of RI and current velocity enables us to evaluate sediment suspension, deposition and transport due to changes in bathymetry of the

shoal, as sediment transport rates are the product of SSC (RI derived from the wave model) and current velocity derived from the hydrodynamic model. It is noted that dynamics and rheology of mud are very different from those of sand (Whitehouse *et al.*, 2000). It is not our intention to do quantitative analysis on mud sediment transport with the RI index, but to evaluate spatial differences on the degree of sediment suspension and deposition due to sand mining. Despite the fact that this paper does not involve a sediment transport model, the processes driving sediment suspension and deposition are discussed and the aforementioned method worked well for a qualitative assessment of sediment dynamics when investigating fine sediment dispersal based on the in-situ data (Kobashi, 2009).

RESULTS

Wave Transformation and Variability of Currents Over the Shoal

Incoming deep-water waves transform as they propagate over complex coastal bathymetry. Spatial differences in wave transformation are similar for all cases, although there are differences in magnitude (Table 4). A particular attention is paid to wave dissipation and refraction for two bathymetries. As for case A1 (Figures 4a,b), substantial wave refraction on the western shoal is clearly evident compared to that without the shoal. As the deep-water wave height becomes lower (case A3, A5 in Figure 4), the difference becomes less evident. On the middle and eastern shoal, the difference in the refraction with and without the shoal is smaller. Despite the fact that the shoal influences wave refraction, it does not seem to significantly affect barrier islands and the region inshore of the shoal except for the western shoal.

The wave height on the western shoal is significantly smaller than that on the eastern shoal (e.g., 50 percent difference between the east and west for case A1 with shoal) (Table 4). With the shoal present, the difference is up to 20 percent higher than the difference with the shoal completely removed for case A1. The spatial difference decreases as the deep-water wave height decreases (Table 4). For cases A4 and A5, the differences in wave height with shoal and without shoal become mostly smaller than 0.1 m. Wave dissipation has a similar trend. Wave dissipation is estimated by the following equation:

$$\{1 - (H_{S[shoal]} / H_{S[deep]})\} \times 100 \quad (2)$$

where, $H_{S[deep]}$ is deep-water wave height listed in Table 2. $H_{S[shoal]}$ is wave height at the locations listed in Table 2. The dissipation decreases from the west toward east for cases A1 and A2 (Table 4). The case A1 has the highest dissipation of all cases. For case A1, the dissipation is approximately 32 percent higher on the western shoal than the eastern flank of the shoal and 50 percent higher than the outside shoal. For cases A3-A5, the east-west trend in dissipation is no longer evident. For all cases, the dissipation outside the shoal is around 10-15 percent. Moreover, the difference in dissipation between the case with shoal and without shoal becomes small though the values of the dissipation are still high. The results suggest that the impacts of changes in bathymetry on wave transformation over the shoal are evident; however, they do not seem to significantly affect the region inshore of the shoal. In addition, for moderate storms to fair weather, influences of the shoal on wave transformation appear to become insignificant.

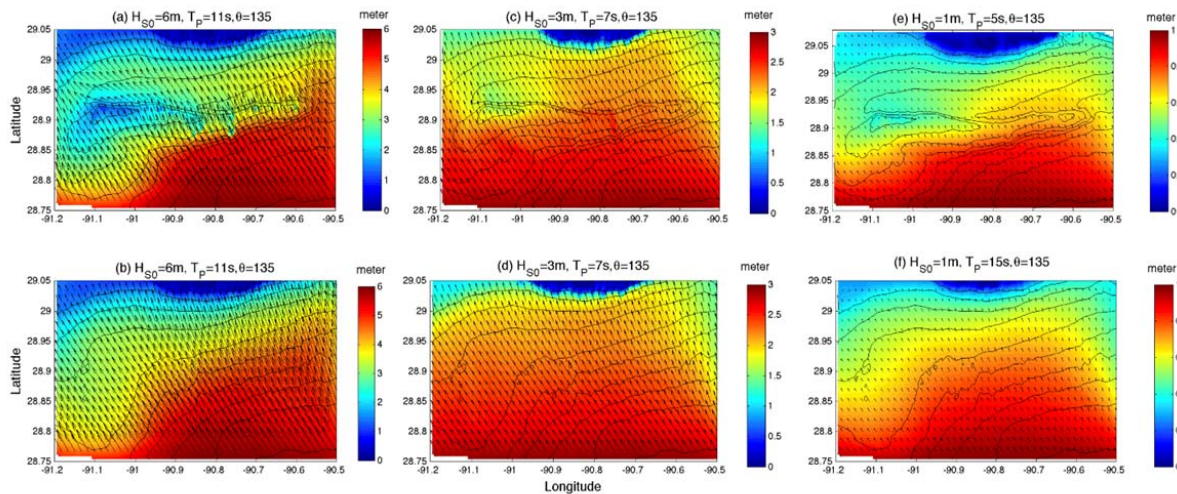


Figure 4. Snapshot of wave height and wave vectors for case A1 (left), A3 (middle), A5 (right); (Top) with shoal and (bottom) without shoal.

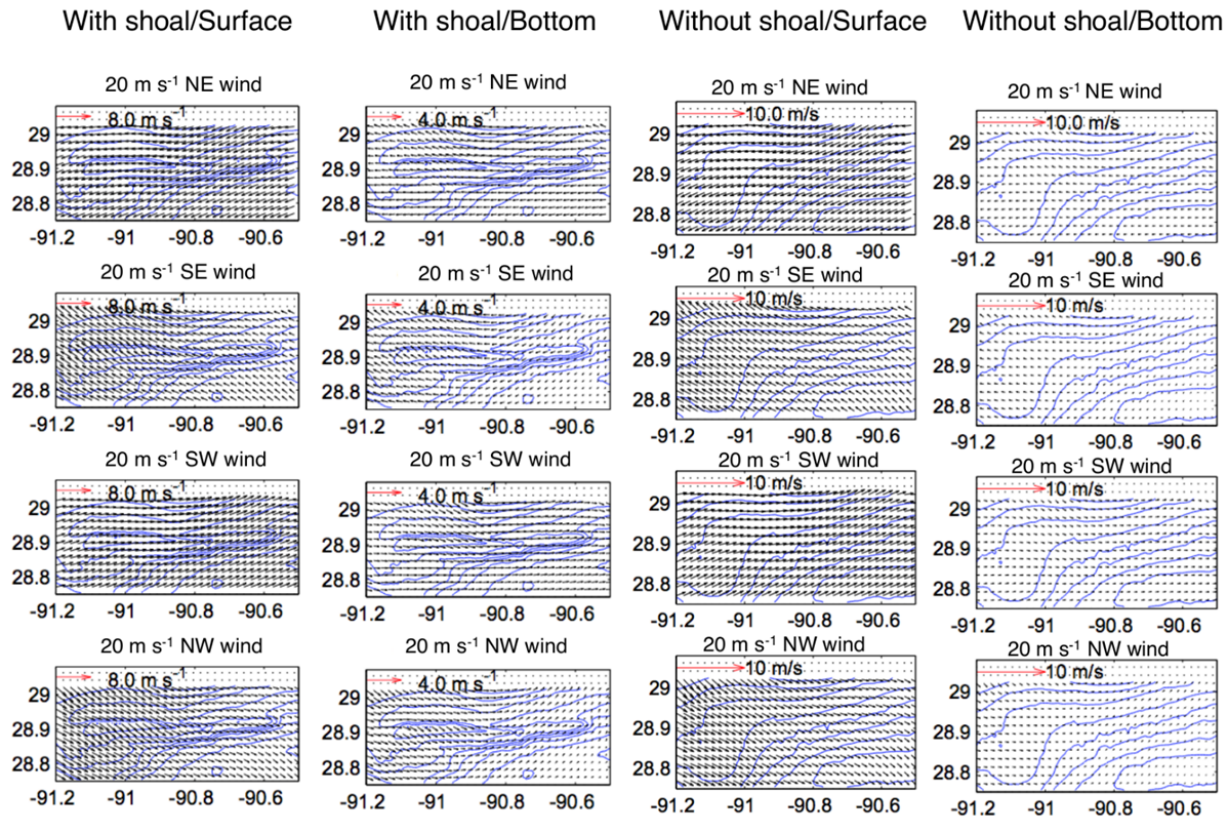


Figure 5. Surface and bottom current vectors for case A (with and without shoal).

When storms pass over the shelf, sea surface slopes and currents respond to a sudden shift in wind fields and subsequently to the geostrophic balance (*cf.* Csanady, 1983). The dependency of current variability with winds is evident for the idealized simulation. As shown in Figure 5, the current pattern is sensitive to the bathymetry and wind direction. Surface currents generally follow the prevailing wind (Figure 5). For instance, during northeasterly wind (NE wind), the eastern shoal experiences southwesterly surface currents and westward currents at the western flank. The surface currents tend to move along isobaths as a result of geostrophic adjustments (*cf.* Csanady, 1983; Swift, 1985). Similar flow characteristics have also been reported over the northern Gulf coasts (*e.g.*, Adams *et al.* (1987) off the southwest pass; Byrnes *et al.* (2004) off the Alabama Coast; Marmorino (1983) over the western Florida shelf). Bottom currents are more variable and are influenced by the shoal bathymetry particularly on the western shoal. The modification of the bottom currents on the western part of the shoal is evident. It is visually evident that both surface and bottom currents are stronger over the shoal than the surrounding shoal because of flow acceleration due to shoal topography to satisfy the continuity, despite increases in bottom friction (Snedden and Dalrymple, 1999; Swift, 1985). The model result suggests that if the shoal did not exist, general spatial patterns of both surface and bottom currents would still

be similar, but be different in magnitude in particular on the western flank of the shoal.

The total volume fluxes are higher for the case with winds from the northeast (NE) and southwest (SW) due possibly to the fact that the currents follow isobaths. Total volume fluxes from the shoal locations with the wind from NE are presented in Figure 6. The volume fluxes are high over the western shoal and decrease from the west to east with decreasing wind speed for the case with shoal. The reduction in volume fluxes from the west to east seems to be linear for all cases regardless of the shoal presence. This trend is same for the result without shoal; however, the volume fluxes are almost always higher when the shoal is present. The difference tends to decrease from the west to east (Figure 6).

Re-suspension and Deposition of Fine Sediments

Sediment suspension and deposition varies with depths and storm conditions as summarized in Table 4 and Figure 7. For energetic storm conditions (*i.e.* case A1 and A2), the RI is high across the domain, but is higher on the middle and eastern shoal than on the western shoal due possibly to higher storm wave dissipation seaward of the western shoal summit. In terms of wind speed and wave height, the cases A1 and A2 account for only 1 percent and 0.03 percent of the time for 2007 and 2008 combined, respectively based on the analysis of WAVCIS CSI-

15 data. As wave energy decreases, as a general trend, the RI on the shoal decreases from the west to east following the change in the shoal bathymetry. For the case A3 (moderate storm), the RI on the western shoal is twice as high as that on the eastern shoal and approximately six times as high as that outside of the shoal. For case A4 (weak storm), the RI is positive on the western and middle shoal, decreases from the west to east, and becomes negative on the eastern flank and outside of the shoal. This suggests that while sediment suspension is expected to occur on the western shoal, the eastern shoal is expected to have sediment deposition. For the case A5 (fair weather), the RI is negative across the domain with the trend similar to case A4, suggesting that no sediment re-suspension is expected during fair weather conditions, which is consistent with the finding of the past studies (e.g., Wright *et al.*, 1997).

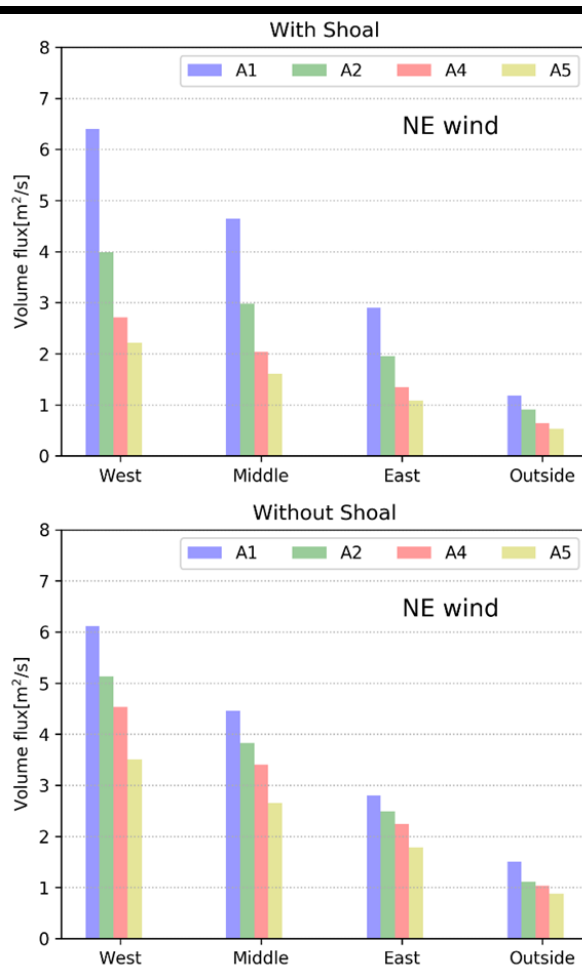


Figure 6. Volume fluxes: (top) with shoal and (bottom) without shoal.

This spatial variability of RI tends to occur when the spatial difference in wave height is small (*i.e.* cases A3, A4 and A5) (Figure 7). When the wave height varies spatially on the shoal

(*i.e.* cases A1 and A2), the difference in RI is small across the shoal. The finding seems to apply regardless of the shoal presence. The result of the RI without the shoal indicates that sediment re-suspension is high on the western shoal during strong storms because of lower wave dissipation rates despite deeper depth (Table 4). However, for most of the model results, the RI without shoal is significantly lower than the RI with shoal, particularly when wave energy is moderate to weak (Table 4). The results indicate that for severe to strong storms, sediments are expected to be re-suspended throughout the shoal; however, for the moderate and weak storms, the sediment suspension has spatial variability with the eastern shoal more prone to sediment deposition.

Table 4. Case study result of each hydrodynamic parameter for case A1-A5. H_s is wave height and RI is re-suspension intensity. The top values in each cell show the model result with shoal and the bottom values show the result without shoal. The dissipation rate is estimated in equation 2.

Parameter	Case	West	Middle	East	Outside
H_s [m]	A1	1.93	2.75	3.85	5.20
		3.32	4.46	4.70	5.17
	A2	2.15	2.91	3.23	3.66
		3.03	3.41	3.29	3.59
	A3	2.01	2.34	2.21	2.55
		2.31	2.35	2.22	2.52
	A4	1.36	1.47	1.38	1.70
		1.43	1.41	1.41	1.67
	A5	0.49	0.64	0.70	0.87
		0.63	0.77	0.73	0.85
Dissipation rate [%]	A1	67.8	54.2	35.8	13.3
		44.7	26.7	21.7	13.8
	A2	46.3	27.3	19.3	8.5
		24.3	14.3	17.8	10.3
	A3	33.0	22.0	26.1	15.0
		23.0	21.7	26.0	16.0
	A4	32.0	26.5	31.0	15.0
		28.5	29.5	29.5	16.5
	A5	51.0	36.0	30.0	13.0
		37.0	23.0	27.0	15.0
RI [$N\ m^{-2}$]	A1	0.97	1.20	1.50	1.39
		1.31	1.42	1.40	1.40
	A2	1.03	1.27	1.18	0.73
		1.13	0.95	0.75	0.78
	A3	0.90	0.92	0.45	0.14
		0.60	0.27	0.10	0.18
	A4	0.16	0.03	-0.14	-0.14
		-0.05	-0.14	-0.18	-0.13
	A5	-0.11	-0.15	-0.19	-0.15
		-0.11	-0.15	-0.19	-0.15

DISCUSSION

Potential impacts of sand mining on physical processes and fine sediment transport on a shore-parallel inner-shelf shoal were investigated by means of numerical modeling. The analysis of the numerical model outputs demonstrates that wave

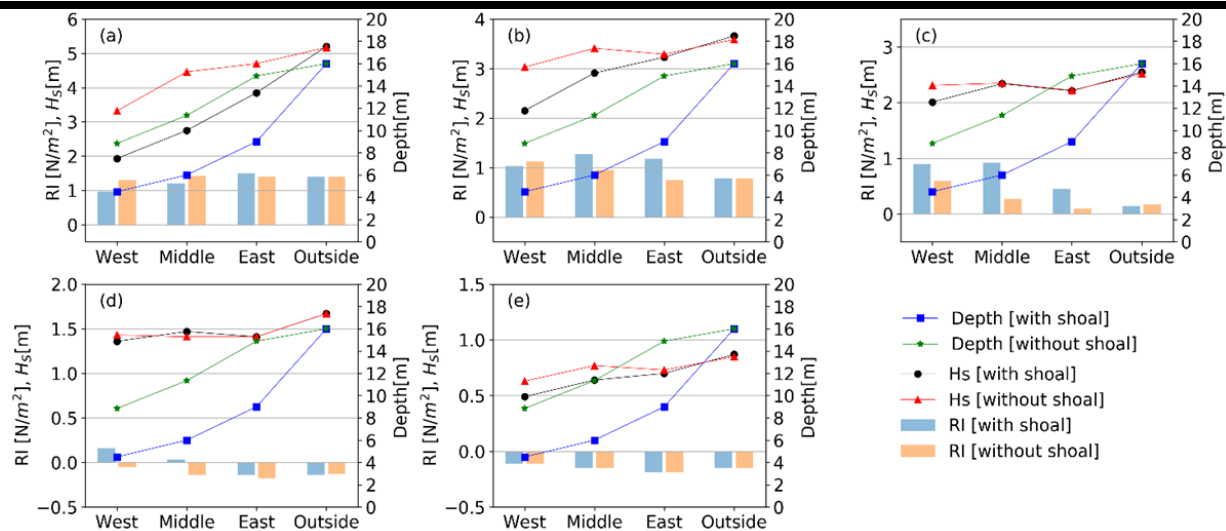


Figure 7. RI on various shoal locations for cases (a) A1, (b) A2, (c) A3, (d) A4, and (e) A5. Water depth and wave height are also included for reference.

dissipation, current flux and sediment suspension and deposition depend strongly on storm conditions and the bathymetry of the transgressive shoal, which is shallowest on the western flank and the deepest along the eastern flank. Wave dissipation, current flux, and sediment re-suspension intensity are all highest on the western flank and decrease toward the eastern flank. This east-west trend creates spatial distribution of sediment suspension and deposition. More specifically, the shallower western shoal is exposed to high sediment transport due to high RI (thus, high SSC) and current flux; whereas, the eastern shoal is prone to the accumulation of fine sediments originated from the Atchafalaya River due to prevailing lower RI and current fluxes. The finding applies to moderate to weak winter storm conditions which are typical winter storms passing over the region. During severe and strong storms, which apply to severe winter storms and tropical cyclones, the entire shoal experiences significant sediment re-suspension and transport. Acoustic backscatter from ADCP, which can be used as a proxy for SSC, at various locations on and around the shoal (see locations in Figure 1) indicates strong sediment suspension during such periods and satellite images support the *in situ* data (Kobashi, 2009). It is reasonable to assume that the western shoal should be exposed to fine sediment plumes more frequently than the eastern shoal due to its proximity to the Atchafalaya River. However, the model result demonstrates just the opposite. The shoal bathymetry appears to play a role in the spatial variability of sediment suspension and deposition since when it happens, wave height is similar across the shoal so that the depth becomes a dominant factor influencing the RI (Figure 7c,d,e). The large-scale sand mining in particular from the western shoal may reduce the spatial variability. Our model result is supported by the fact that the sediments sampled in the past on the western shoal were sandy and those sampled on the middle and eastern shoal were a mixture of sand and mud during the winter-storm seasons (Pepper and Stone, 2004; Kobashi, 2009). It is noted that the repeated sampling of shoal sediments indicates that the area of

the fine sediment accumulation on the middle and eastern shoal is likely patchy and temporal, possibly due to rough surface of the shoal sandy bottom and frequent exposure to winter storms (Kobashi, 2009).

The findings provide an insight as to how atmospheric forcing, fluvial fine sediments, hydrodynamics and shoal bathymetry come into play on sedimentary environment of the shoal in a complex manner. The paper also provides important knowledge on how changes in bathymetry could influence morpho-hydrodynamics of the shoal. Large-scale mining will likely contribute to changes in wave dissipation, current fluxes, and sediment suspension, deposition and transport rates as a direct consequence of deepening the shoal bathymetry. The mining from the shallower western shoal may reduce spatial variability of morpho-hydrodynamic processes. Further, substantial deepening of the shoal may have a negative impact on benthic habitats by exposing them to fine sediment accumulation more frequently, particularly during the spring flood season (Grippo *et al.*, 2009; Kobashi, 2009). The assessment of the small-scale targeted sand mining, which was conducted as a separate study earlier, demonstrated that small-scale mining did not appear to have a significant change in morpho-hydrodynamic parameters of the shoal (Kobashi, 2009). However, excessive deepening of the shoal even for small-scale mining may inhibit benthic primary production by potentially enhancing accumulation of fine sediments and inhibiting light penetration, which are reported to be detrimental for benthos (Grippo *et al.*, 2009). In order to minimize the impacts, those potential changes should be taken into account. The dredging from the deeper eastern flank of the shoal (e.g., area A in Figure 1) may result in lesser impacts on morpho-hydrodynamics on the shoal than the shallower middle and western portions of the shoal. Overall, the findings from this study are consistent with the field measurements (Kobashi, 2009; Kobashi *et al.*, 2007) and a collaborative biological study of the shoal undertaken in the past (Stone *et al.*, 2009).

SUMMARY

A coastal modeling suite was implemented to investigate changes in hydrodynamics and consequent sedimentary environment of an inner-shelf shoal on which fine sediments are covered. The model outputs and their analyses led to the following conclusions. For the hypothetical modeling scenario with the complete removal of the shoal, waves and currents would undergo significant modifications. Current variability simulated for different spatial locations shows a similar trend for with and without the shoal. However, they differ in magnitude for the two bathymetric configurations. There is an obvious east-west trend in morpho-hydrodynamic processes depending on storm conditions and the eastern part of the shoal is prone to accumulating fine-grained sediments during moderate and weak storms. The data presented in this paper suggests that large-scale sand mining may significantly change morpho-hydrodynamics of the shoal, particularly waves, associated wave-induced sediment re-suspension, magnitude in current flux, and consequent sediment fluxes. Also, the large-scale dredging may result in enhanced accumulation of fine sediments, which could adversely affect benthic habitats of the shoal. Thus, we should take into account such potential impacts before implementing any large-scale dredging of the shoal. Even though small-scale dredging does not appear to have a significant change in morpho-hydrodynamic parameters, excess deepening of bathymetry raises a concern on sediment deposition. The dredging from the deeper eastern flank of the shoal may result in lesser impacts on hydrodynamics on the shoal than the shallower middle and western portions of the shoal. Overall, the findings are consistent with the field measurements and a collaborative biological study of the shoal.

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